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Applying reinforcement learning in Bitcoin trading to select technical strategies based on Deep Q-Network

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ABSTRACT

This paper proposes a Deep Q-Network-based reinforcement learning framework to optimize the selection of technical trading strategies for Bitcoin. Instead of directly predicting prices, the agent chooses among five predefined strategies—RSI, SMA Crossover, Bollinger Bands, Momentum-20d, and VWAP Reversion—allowing actions to remain interpretable while adapting dynamically to market conditions. Market states are constructed from technical indicators, normalized, and reduced in dimensionality using Principal Component Analysis to ensure robustness and efficiency. Empirical tests on Bitcoin data from 2022 to mid-2025 show remarkable performance: starting from \$1,000,000, the agent achieved more than 120-fold growth in Net Asset Value, significantly outperforming Buy-and-Hold and single-strategy benchmarks. The model flexibly emphasized momentum during uptrends, limited mean-reversion during breakouts, and managed drawdowns when temporarily misaligned. Out-of-sample results further confirmed generalization. Overall, the framework demonstrates a practical, explainable, and scalable approach to algorithmic trading in digital asset markets.

IMPACT STATEMENT

This study proposes a reinforcement learning framework that selects among predefined technical strategies for Bitcoin trading, improving interpretability and reducing model instability. Using Deep Q-Networks with PCA-based features, the agent achieves over 120-fold NAV growth and shows strong adaptability across different market regimes. The findings demonstrate a practical and scalable approach to algorithmic trading in highly volatile digital asset markets.

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1. Introduction

In recent years, Bitcoin has emerged as a prominent digital asset, disrupting traditional financial systems and garnering substantial interest from both institutional and individual investors (Hossaion et al., 2023). As the inaugural cryptocurrency and the one with the highest market capitalization, Bitcoin exhibits unique attributes compared to conventional assets, including decentralization, finite supply, and elevated price volatility. These distinctive features present both opportunities and challenges for investors, underscoring the necessity for research into effective Bitcoin investment strategies.

Unlike traditional financial instruments such as stocks or bonds, Bitcoin operates on a decentralized blockchain network without endorsement from any government or central authority (Vergne, 2020). This decentralization introduces novel dynamics in price fluctuations and investor behavior, rendering traditional pricing and risk assessment models less applicable. Consequently, there is a pressing need to develop tailored investment strategies that accommodate the idiosyncratic characteristics of the cryptocurrency market.

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Moreover, Bitcoin's pronounced volatility represents a double-edged sword, offering potential for substantial returns while simultaneously entailing significant risk of loss. Investors face unpredictable price swings influenced by macroeconomic factors, regulatory developments, market sentiment, or even social media trends. To mitigate these risks, researchers must investigate portfolio management techniques, hedging instruments, and algorithmic trading strategies capable of adapting to dynamic market conditions.

Furthermore, the rapid evolution of the cryptocurrency ecosystem—including derivatives, decentralized finance platforms, and institutional investment vehicles—further complicates investment decision-making. As markets grow more sophisticated, the demand intensifies for data-driven approaches and advanced technologies, such as machine learning, reinforcement learning, and big data analytics. These methodologies can facilitate trend identification, optimize asset allocation, and enhance timing decisions in highly volatile environments.

An additional rationale for further research stems from the scarcity of historical data and regulatory ambiguity. Bitcoin remains a relatively nascent asset, with fewer than two decades of market history. Concurrently, regulatory frameworks continue to evolve and differ across jurisdictions, generating legal uncertainties that influence investment behavior. Scholarly and applied research can address these deficiencies by offering insights into long-term market patterns, investor sentiment, and the ramifications of varying policies.

Bitcoin's price was virtually negligible during its formative years. From its inception in January 2009 until mid-2010, BTC traded at fractions of a cent or even less, initially lacking any established market value (Malone, 2019). In October 2009, the New Liberty Standard set an exchange rate of $\$1 = 1,309.03$ BTC (approximately $\$0.00076$ per BTC) based on electricity costs (Standard, 2009). The first formalized market emerged in March 2010 on BitcoinMarket.com, where BTC was valued at around $\$0.003$ (Sharad Mangrulkar & Vijay Chavan, 2024). By July 2010, the price had ascended to approximately $\$0.08$ amid increasing adoption (Marinescu, 2018). Throughout 2011, Bitcoin surpassed the $\$1$ threshold in February and peaked at around $\$31$ in June before undergoing a correction—an extraordinary appreciation from near-zero value within just two years (Sembler, 2023).

Figure 1 illustrates BTC trading below $\$1,000$ until mid-2017, with prices surging above $\$18,000$ in December 2017, driven by speculative fervor. A precipitous correction in early 2018 depressed prices below $\$7,000$, succeeded by a recovery phase that culminated near $\$64,000$ in April 2021. Market turbulence in 2022 reduced Bitcoin's value to approximately $\$18,000$, yet a resurgence in 2023 and early 2024 elevated prices beyond $\$65,000$. More recently, Bitcoin achieved an all-time high exceeding $\$100,000$, indicative of robust demand and extensive adoption.

As Bitcoin gains broader acceptance and integrates into diversified investment portfolios, comprehending its interactions with other asset classes becomes imperative. Extensive research has elucidated Bitcoin's potential as a diversification tool, a hedge against risk, and a safeguard against inflation. These findings hold particular relevance for financial institutions, asset managers, and individual investors seeking to incorporate Bitcoin into their portfolios.



Figure 1. BTC closing price by day.

Numerous Bitcoin investment strategies exist, each tailored to varying risk appetites, investment horizons, and market dynamics. Several prominent strategies are delineated below.

First, the Buy-and-Hold strategy represents one of the most established and straightforward approaches in finance (Dichtl, 2020). Under this paradigm, investors acquire an asset—such as equities, gold, or Bitcoin—and retain it over an extended period, irrespective of short-term price volatility. The underlying premise posits confidence in the asset's long-term appreciation, thereby maximizing returns through minimized trading activity and avoidance of transient market oscillations. This strategy's primary merits include operational simplicity, reduced market monitoring requirements, and diminished transaction costs. In protracted bullish markets, such as Bitcoin's trajectory from 2013 to 2021, Buy-and-Hold yields substantial gains. Nonetheless, its limitations manifest in the absence of protective mechanisms during market reversals or protracted downturns. In environments characterized by high volatility or lateral movement, adherents may endure significant drawdowns and forego short-term opportunities. Accordingly, this approach suits long-term investors possessing high risk tolerance and patience.

The trend-following strategy constitutes a prevalent technique in technical analysis, predicated on the notion that prices exhibit directional persistence over time (Liu, 2023). Practitioners typically employ indicators such as simple moving averages (SMA), exponential moving averages (EMA), the Moving Average Convergence Divergence (MACD), or the Average Directional Index (ADX) to identify and validate trends—for instance, initiating purchases upon a golden cross (short-term SMA surpassing the long-term SMA) and divesting during a death cross (the inverse). The strategy's objective is to align with prevailing momentum rather than anticipate reversals, thereby capitalizing on sustained bullish or bearish phases. Its salient advantages encompass conceptual accessibility, ease of execution, and efficacy amid pronounced market trends. However, notable drawbacks arise in range-bound or erratic markets, where spurious signals may precipitate successive losses. Moreover, the inherent lag in trend indicators often results in delayed entry and premature exit from trends.

The mean-reversion strategy constitutes a trading approach predicated on the premise that asset prices oscillate around a long-term equilibrium, with substantial deviations from this mean being transient (Jansen, 2020). Consequently, when prices escalate excessively above the mean, investors anticipate a reversion downward; conversely, when prices decline unduly below the mean, a rebound is expected. Commonly employed indicators for ascertaining overbought or oversold states encompass Bollinger Bands, the Relative Strength Index (RSI), and measures of moving average divergence. This strategy's principal advantage lies in its capacity to yield consistent returns in range-bound markets, wherein prices lack a definitive trend but frequently rebound upon encountering support or resistance levels. Nevertheless, a notable limitation arises when the market breaches the equilibrium range and establishes a robust trend, potentially incurring substantial losses through persistent attempts at 'bottom fishing' or 'top fishing' at inopportune moments. Thus, practitioners of this strategy must implement robust risk management protocols and accurately discern suitable market regimes.

The indicator-based strategy represents a trading methodology that leverages mathematical computations derived from historical price and volume data to inform decisions regarding asset acquisition, divestiture, or retention (Lecegui et al., 2022). Prominent indicators include the RSI, MACD, Stochastic Oscillator, Volume Weighted Average Price (VWAP), and assorted moving averages. Each indicator typically elucidates a particular market facet, such as momentum, trend direction, volatility, or overbought/oversold conditions. The strategy's merits encompass its accessibility, amenability to automation, and widespread integration within trading platforms. Investors may construct rule-based systems, such as initiating purchases when the RSI falls below 30 concurrent with a price crossover above a moving average, or divesting upon a MACD signal line crossover. However, a primary drawback pertains to the indicators' inherent lag and vulnerability to noise amid heightened volatility or trendless periods. Moreover, excessive reliance on multiple indicators may engender conflicting signals, thereby diminishing efficacy absent rigorous optimization and backtesting.

Supervised machine learning in financial trading entails the application of models to forecast price trajectories or market trends utilizing labeled historical datasets (Jansen, 2020). Within this paradigm, the target variable—or 'label'—typically encompasses price direction (upward or downward), anticipated returns, or the likelihood of specific financial occurrences. Frequently utilized algorithms comprise Support Vector Machines (SVM), Random Forests, XGBoost, and neural architectures such as Long Short-Term Memory (LSTM) networks or Multi-layer Perceptrons (MLP). Practitioners train these models on feature sets including RSI values, simple moving averages, logarithmic returns, and trading volumes, enabling the identification of lucrative market configurations. This approach's strengths include its proficiency in discerning intricate nonlinear relationships and processing voluminous datasets. Nonetheless,

limitations manifest in the models' propensity for short-term predictive accuracy alone, without direct optimization for cumulative returns, and an absence of sequential decision modeling akin to reinforcement learning. Furthermore, supervised models are susceptible to overfitting absent meticulous hyperparameter tuning or when training data inadequately captures diverse market scenarios.

Each methodology exhibits efficacy predominantly under specific market conditions: Buy-and-Hold excels in protracted trends, trend-following thrives amid pronounced momentum, mean-reversion performs well in sideways markets, and supervised learning suits short-term prognostication. However, financial markets are inherently dynamic, transitioning fluidly between trending, ranging, and breakout phases. A singular, static strategy proves inadequate for encompassing all contingencies and remains susceptible to underperformance in unfavorable regimes.

This necessitates the development of a hybrid strategy that amalgamates the strengths of individual approaches while dynamically alternating among them contingent upon prevailing market states. Such a framework facilitates the exploitation of trend-following during unambiguous directional movements, transitions to mean-reversion amid narrow oscillations, integration of supervised signals for enhanced short-term precision, and adherence to Buy-and-Hold in extended low-volatility intervals.

In the realm of cryptocurrency investment, particularly Bitcoin trading, reinforcement learning has garnered escalating attention as a prospective instrument for constructing adaptive trading systems (Fang et al., 2022). Conventional machine learning techniques frequently emphasize price direction prediction or future value estimation; however, these are constrained by data noise, market volatility, and overfitting risks. An innovative paradigm to surmount these limitations involves deploying reinforcement learning (RL) to cultivate an agent that eschews direct generation of buy or sell directives based on prices, opting instead to select among predefined technical trading strategies.

This architecture streamlines the decision space and more authentically mirrors traders' practices, who commonly draw upon established methodologies such as moving average crossovers, RSI thresholds, Bollinger Bands, or MACD signals. These strategies encapsulate domain-specific expertise attuned to particular market contexts. Rather than devising novel tactics ab initio, the RL agent functions as an overarching selector, tasked with designating the optimal strategy at each juncture.

The learning milieu is typically formalized as a Markov decision process (Boucherie & Van Dijk, 2017), wherein the state vector incorporates attributes such as market volatility, trend metrics, momentum indicators, or macroeconomic cues. The action space comprises not discrete buy/sell/hold commands, but rather the repertoire of available technical strategies—for instance, electing 'RSI trading,' 'MACD strategy,' or 'resistance breakout strategy.' Subsequently, each selected strategy autonomously yields buy/sell signals per its intrinsic rules, with the agent accruing rewards predicated on portfolio metrics, including realized profits, Sharpe ratio, or maximum drawdown.

This approach offers several advantages. Foremost, it enhances explainability, as the agent's decisions can be interpreted through established strategies rather than inscrutable machine learning heuristics. Secondly, it mitigates the complexity of the learning process, inasmuch as the agent eschews direct mapping of price data to actions, instead selecting from a discrete ensemble of expert strategies. Thirdly, it bolsters adaptability, enabling the agent to modulate its strategy in response to market vicissitudes.

To train the agent, algorithms such as Deep Q-Networks (Brim, 2020), Proximal Policy Optimization (Wu et al., 2021), or Advantage Actor-Critic (Liu et al., 2020) may be employed. Historical market data serves as the training milieu, with reward functions tunable to emphasize risk-adjusted metrics. Over iterative epochs, the agent ascertains optimal strategy alignments with prevailing market regimes, thereby progressively refining its decision-making efficacy.

Employing reinforcement learning for strategy selection, rather than direct signal generation, constitutes a robust and pragmatic methodology within the financial domain. This paradigm synergistically integrates domain expertise with artificial intelligence's adaptive capabilities, furnishing a versatile scaffold for developing trading agents in intricate markets such as Bitcoin.

1.1. Academic contribution and research gap

Although numerous prior investigations have leveraged reinforcement learning in financial trading by directly outputting buy, sell, or hold directives, such methodologies frequently exhibit erratic performance

and susceptibility to overfitting, attributable to the cryptocurrency market's pronounced volatility and non-stationarity. The extant research lacuna pertains to the paucity of models that embed domain knowledge within the action space to augment learning stability and interpretability.

This study redresses that deficiency by proposing an innovative reinforcement learning framework wherein the action space comprises a curated suite of technical trading strategies—such as RSI, SMA crossovers, momentum indicators, VWAP, and Bollinger Bands—rather than granular trading actions. Within this configuration, the RL agent functions as a meta-level arbiter, acquiring proficiency in designating the most apposite strategy amid fluctuating market conditions, predicated on a parsimonious state representation derived from normalized technical indicators and dimensionality-reduced via principal component analysis (PCA).

The principal scholarly contribution of this inquiry resides in the reconceptualization of the reinforcement learning paradigm as a meta-strategy selection endeavor, which concurrently streamlines decision processes and more faithfully emulates empirical trading practices. This architecture augments model transparency, attenuates overfitting hazards, and facilitates superior acclimation to market regime shifts. Empirical evaluations utilizing Bitcoin data spanning 2022 to 2025 evince enhanced performance relative to conventional benchmarks, thereby substantiating the framework's theoretical merit and operational viability in automated digital asset trading.

2. Literature review

2.1. Overview of Bitcoin investment strategies

Research on Bitcoin investment strategies has proliferated substantially over the past decade, mirroring the swift maturation of the cryptocurrency market. These strategies can broadly be categorized into several typologies, encompassing traditional long-term investment paradigms as well as technical and machine learning-driven methodologies.

As financial markets exhibit escalating volatility and prognostic challenges, numerous investors resort to conventional investment strategies to augment returns while mitigating risks. Among the most prevalent are the buy-and-hold (HODL), dollar-cost averaging (DCA), and portfolio allocation approaches (Kapalczynski & Lien, 2021). Although each operates on distinct principles, they converge on the objective of establishing durable and resilient investment architecture adaptable to varied market regimes and investor risk appetites.

The buy-and-hold strategy is premised on the conviction that assets like Bitcoin will appreciate in value over the long term, notwithstanding transient volatility (Jaquart et al., 2021). Eschewing attempts to time the market through opportunistic buying and selling, proponents acquire and retain assets across protracted durations, positing that secular uptrends will ultimately offset interim drawdowns. This approach garners particular favor within cryptocurrency domains, where price trajectories frequently manifest pronounced bullish cycles over extended horizons.

Conversely, dollar-cost averaging seeks to attenuate risk by dispersing capital infusions temporally, rather than deploying a singular lump sum. Through periodic commitments of fixed sums—be they weekly, monthly, or quarterly—this methodology diminishes the probability of market entry at zeniths and averages the acquisition cost basis longitudinally. It cultivates disciplined investment behavior, proving especially efficacious in environments of acute volatility, such as cryptocurrency markets.

Portfolio allocation constitutes another pivotal strategy emphasizing diversification. It entails apportioning capital across heterogeneous asset classes, including equities, fixed-income securities, commodities, and digital currencies, to equilibrate returns and curtail idiosyncratic risks (Abbas & Usman, 2025). Empirical inquiries have demonstrated Bitcoin's propensity for low or negative correlations with orthodox financial instruments, such as equities and bonds (Mukherji, 2022). This attribute renders it a compelling adjunct for portfolio diversification, particularly amid a globalized investment milieu characterized by intensifying inter-asset interdependencies.

Although conventional strategies like HODL, DCA, and portfolio diversification confer salient advantages, their applicability hinges on investors' fiscal aspirations, temporal scopes, and risk thresholds. As digital assets progressively assimilate into mainstream finance, comprehending and adaptively

implementing these strategies emerges as instrumental to optimizing performance and enduring market perturbations.

Beyond long-term orientations, a burgeoning cohort of investors employs technical analysis to capitalize on ephemeral price oscillations (Ayala et al., 2021). Technical analysis harnesses historical price and volume datasets to derive buy or sell signals via indicators such as the Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Bollinger Bands, and moving averages including the Simple Moving Average and Exponential Moving Average (EMA).

Each indicator proffers discernment into discrete market facets. For instance, the RSI gauges momentum and delineates overbought or oversold states—thresholds exceeding 70 denote overbought conditions, whereas those below 30 indicate oversold. The MACD elucidates interrelations between dual moving averages, facilitating the detection of prospective trend inversions and momentum divergences. Bollinger Bands evaluate volatility and pinpoint potential breakout junctures, while SMA and EMA serve to denoise price series and accentuate dominant trends, with the EMA exhibiting heightened responsiveness to contemporaneous price alterations.

Technical trading strategies frequently amalgamate these indicators into codified protocols, such as acquiring positions when the RSI registers low values concurrent with prices breaching the inferior Bollinger Band, or divesting upon the MACD traversing beneath its signal line. A key merit of technical strategies resides in their congruence with algorithmic trading and automated systems, enabling mechanization and instantaneous market reactivity.

Nevertheless, technical strategies evince conspicuous constraints. As evidenced in empirical analyses (Malone, 2019), although select indicator-centric methodologies may surpass stochastic trading—particularly in directional markets—their efficacy remains profoundly conditional upon ambient market states. In lateral or subdued-volatility regimes, these approaches are susceptible to spurious signals, amplifying loss probabilities absent efficacious management.

Notwithstanding their utility, technical indicators warrant non-dogmatic application. Investors ought to calibrate strategies in consonance with evolving market milieus and incorporate macroeconomic or fundamental analyses to foster a holistic decisional framework.

In recent years, machine learning modalities have attained prominence in financial trading, inclusive of the Bitcoin arena. Architectures such as Support Vector Machines, Random Forests, and neural networks have been marshaled to prognosticate price trajectories and emit signals (Kurani et al., 2023). Nonetheless, owing to Bitcoin data's pronounced volatility and nonlinear constitution, accurate prognostication persists as arduous. Overfitting and extramural predictive instability constitute recurrent impediments, especially within expeditiously mutating markets.

Certain scholars have additionally scrutinized arbitrage and signal-oriented algorithmic trading across disparate cryptocurrency venues. Although these modalities may afford high-frequency remunerative prospects, they necessitate sophisticated infrastructure, minimal-latency fulfillment, and synchronous access to inter-exchange order books—prerequisites seldom attainable by retail investors.

The scholarly corpus invariably underscores that, while Bitcoin proffers extraordinary remunerative prospects, it concomitantly entails formidable risks. Ergo, electing a germane investment strategy must be conjoined with rigorous risk appraisal and adaptive implementation apparatuses attuned to the exigencies of a profoundly kinetic and speculative marketplace.

2.2. Literature review on reinforcement learning

Reinforcement learning is an important and growing field in machine learning that studies how an agent can learn to make optimal decisions through continuous interaction with the environment (Padakandla, 2022). The core goal of reinforcement learning is to maximize the total cumulative reward an agent receives over time by discovering and exploiting the most efficient actions in each situation.

A reinforcement learning system is typically modeled using a Markov Decision Process—a standard mathematical construct for describing decision-making environments with a stochastic component. An MDP consists of the following main components: states—representing the current situation; actions—the choices the agent can make; rewards—feedback signals about the quality of the action; and

transition probabilities—describing the probability of the environment moving from one state to another after each action.

During learning, the agent uses a policy—a behavioral strategy that determines what action to take at each state. The policy can be a probability or a function that directly determines the action (Zhu et al., 2024). At the same time, the agent constructs a value function, which estimates the expected reward it can achieve when starting from a certain state (or state-action pair) and acting according to the current policy.

Reinforcement learning encompasses many algorithms with different approaches. Basic algorithms such as Q-Learning and SARSA use a value table to update the estimate of the function Q —which represents the value of each action at each state. They fall into the category of off-policy or on-policy, value-based learning and use an ε -greedy strategy to balance exploitation and exploration.

With the development of deep learning, Deep Q-Network has extended the capabilities of Q-Learning by using deep neural networks to approximate the Q function, making it possible to apply reinforcement learning to large and continuous state spaces, such as in Atari games (Brim, 2020). In addition, policy optimization-based algorithms such as Policy Gradient, Proximal Policy Optimization, Advantage Actor-Critic or Deep Deterministic Policy Gradient have been developed to directly update policies, which are capable of handling continuous action problems and complex environments well (Liu et al., 2020; Wu et al., 2021).

Reinforcement learning has not only been successful in traditional fields such as robotics and video games (Atari, AlphaGo), but has also gradually asserted its role in practical applications such as communication network optimization, energy management, smart transportation, and especially in finance, where sequential decision making and highly uncertain environments are typical features. Recently, reinforcement learning has been applied to automated financial trading, asset allocation, and portfolio optimization, opening up new directions for intelligent trading strategies based on real-time data and market feedback.

2.3. Literature review on reinforcement learning in Bitcoin investment

The application of reinforcement learning in financial investment, particularly for highly volatile assets such as Bitcoin, is emerging as a dynamic and promising research direction. Due to its capacity to learn from interaction and adapt to complex, uncertain environments, RL enables the development of intelligent trading systems that can respond effectively to ever-changing market conditions.

An early and widely studied approach involves training the RL agent to directly learn trading actions—specifically, whether to buy, sell, or hold—based on observed price series or derived technical indicators such as the Relative Strength Index and Moving Average Convergence Divergence. For example, research by Mondol (2024) applied Deep Q-Network and Deep Deterministic Policy Gradient algorithms to construct an automated trading system tailored to cryptocurrency markets. However, financial markets—especially those involving digital assets—are notoriously noisy, non-stationary, and heavily influenced by external factors such as macroeconomic news and investor sentiment. These characteristics make RL models highly susceptible to overfitting and limit their ability to generalize to unseen data.

To address these limitations, a newer framework has been introduced in which the RL agent does not learn trading actions directly but instead selects among a predefined set of technical trading strategies. This abstraction reduces the decision space and mirrors real-world trading behavior, where investors often operate based on established strategies rather than raw market signals.

Another important consideration in applying RL to trading is the use of appropriate performance evaluation metrics. In addition to absolute returns, models are increasingly assessed using financial risk-adjusted indicators such as the **Sharpe Ratio**, which measures excess return per unit of risk; the **Sortino Ratio**, which focuses specifically on downside volatility; and **Maximum Drawdown**, which captures the largest decline from a peak to a trough. Recent studies have begun to explore **multi-objective reinforcement learning**, where the agent simultaneously optimizes for multiple financial metrics. This approach aims to deliver more robust and stable performance across varying market regimes.

Furthermore, the integration of RL with deep learning techniques is gaining traction. Deep neural networks such as Long Short-Term Memory (LSTM) and Transformer architectures are employed to process sequential price data and extract latent features that inform trading decisions. While these architectures provide powerful tools for capturing temporal dependencies and nonlinear patterns, they also introduce new challenges. These include heightened sensitivity to abrupt market shifts, data instability, and the significant impact of transaction costs on profitability.

To mitigate such challenges, some research efforts have focused on training RL agents within simulated or synthetic environments. These controlled settings allow the model to explore a wide range of scenarios, improve its generalization ability, and test its robustness under stress conditions before deployment in live markets.

2.4. Theoretical basis

This study is grounded in an interdisciplinary integration of financial economics, behavioral finance, market microstructure theory, and machine learning—particularly reinforcement learning—to address the complex and dynamic nature of Bitcoin trading.

1. Efficient market hypothesis and its limitations

The Efficient Market Hypothesis (EMH) posits that asset prices fully reflect all available information, implying that it is impossible to consistently outperform the market through any trading strategy (Malkiel, 1989). However, the Bitcoin market—characterized by high volatility, speculative cycles, and structural inefficiencies—presents significant challenges to this assumption. Empirical observations of price anomalies, irrational spikes, and frequent mispricings suggest the presence of exploitable inefficiencies. These deviations from equilibrium pricing create a fertile environment for adaptive algorithmic approaches, such as reinforcement learning, to discover patterns that conventional models may fail to capture.

2. Behavioral finance and psychological drivers

In contrast to EMH, behavioral finance emphasizes the influence of cognitive biases, heuristics, and emotional responses on investor decisions and asset pricing (Gabhane et al., 2023). Cryptocurrency markets are particularly vulnerable to behavioral phenomena such as overreaction, underreaction, loss aversion, and herd behavior. These dynamics are amplified by social media, online communities, and speculative sentiment, often leading to extreme price swings disconnected from fundamental value. Such conditions justify the use of trading strategies built on momentum and mean-reversion principles, which aim to systematically exploit predictable departures from rational price behavior.

3. Market microstructure and execution efficiency

Market microstructure theory investigates how trading mechanisms, order types, liquidity, and transaction costs influence price formation and market behavior (Lehalle & Laruelle, 2018). In Bitcoin markets—where liquidity is fragmented across exchanges and order book depth fluctuates—efficient trade execution becomes a critical consideration. Execution strategies such as Volume Weighted Average Price and other liquidity-sensitive methods are designed to reduce slippage and market impact by aligning with prevailing order flow and minimizing exposure to adverse selection. Incorporating these considerations into algorithmic systems enhances both profitability and practical feasibility in real-world trading environments.

4. Reinforcement learning for sequential decision-making

From a computational perspective, reinforcement learning provides a formal framework for learning optimal policies through interaction with an uncertain environment (Wang et al., 2021). Financial trading is inherently a sequential decision-making process where the agent must balance exploration of new strategies and exploitation of known profitable ones. Recent advances in deep reinforcement learning allow agents to approximate optimal behaviors even in high-dimensional, non-linear spaces with delayed rewards. This makes RL particularly suitable for cryptocurrency trading, where environments are dynamic, noisy, and prone to regime shifts.

This study employs a meta-strategy approach in which the RL agent selects among a discrete set of predefined trading strategies rather than issuing individual trade actions. This design reduces the

complexity of the action space while incorporating expert knowledge into the learning process. It also improves interpretability and robustness, which are often critical in high-stakes financial applications.

5. Adaptive market perspective

Moving beyond the static assumptions of EMH, the adaptive market perspective conceptualizes financial markets as evolving ecosystems, where strategy effectiveness depends on environmental shifts, competition, and agent learning capacity (Lo, 2004). Under this paradigm, market efficiency is not a fixed condition but a dynamic state that emerges from the interaction of market participants. Consequently, a successful trading system must be capable of adjusting its behavior in response to changing conditions. Reinforcement learning aligns naturally with this view by enabling agents to continually update policies through feedback, improving their adaptability across various market regimes.

2. Method

This study develops a systematic framework that integrates financial data preprocessing, feature engineering, and reinforcement learning techniques to construct an intelligent trading agent. The methodology is structured into four main components. First, raw Bitcoin price data is transformed through the calculation of technical indicators, followed by normalization and dimensionality reduction to produce a compact yet informative feature set. Next, a tailored Feature Selection Reinforcement Learning (FSRL) environment is designed to capture the dynamics of the trading process. Building on this environment, the reinforcement learning problem is formally defined, including state representation, action space, and reward function. Finally, the Deep Q-Network (DQN) algorithm is implemented as the core learning mechanism, enabling the agent to approximate optimal policies and adapt to evolving market conditions.

As the first step of the framework, **data preprocessing** plays a critical role in shaping the quality and interpretability of the information supplied to the learning agent. Since raw Bitcoin prices are highly volatile and non-stationary, direct usage would hinder the model's ability to capture meaningful patterns. Therefore, the data is systematically transformed into a set of well-defined technical indicators, normalized to ensure comparability across scales, and further refined using dimensionality reduction techniques. This process ensures that the subsequent reinforcement learning environment is grounded in structured, noise-reduced, and information-rich features, providing a robust foundation for the learning tasks that follow.

3.1. Data preprocessing

Data preprocessing is an essential step in constructing the input features for the reinforcement learning framework. The raw daily closing prices of Bitcoin contain valuable information but are inherently noisy, non-stationary, and difficult to interpret directly. To overcome these limitations, the preprocessing stage transforms the raw data into structured financial indicators that better reflect underlying market dynamics. This transformation serves two purposes: first, it highlights salient features such as trend, momentum, and volatility; second, it standardizes and compresses the data to make it more suitable for statistical learning.

The preprocessing procedure is divided into two main tasks. In the first stage, a set of widely recognized technical indicators is computed from the daily closing price series, providing the agent with interpretable signals commonly used in financial analysis. In the second stage, these features are normalized and subjected to dimensionality reduction through Principal Component Analysis, which removes redundancy and reduces noise while preserving the essential information. Together, these steps ensure that the learning model operates on a clean, compact, and information-rich feature set.

3.1.1. Calculation of technical indicators

During the process of characterizing the input data of the reinforcement learning agent, several important technical indicators are calculated from the daily closing price series of Bitcoin. Specifically, the closing price at time t is denoted by C_t . To obtain a stationary, scale-aware measure of price variation, we compute the log-return, a continuous form of return, defined as the difference of the natural logarithms between today's closing price and the previous day's:

$$r_t = \log C_t - \log C_{t-1} \quad (1)$$

This transformation linearizes multiplicative changes, normalizes returns across different price levels, and typically yields distributions more amenable to machine learning models.

To capture momentum and potential regime shifts, we incorporate the Relative Strength Index, a well-established oscillator that reflects overbought or oversold market conditions. We begin by computing the average positive profit (gain) and average loss over the last 14 sessions as follows:

$$\text{gain}_t = \frac{1}{14} \sum_{i=0}^{13} \max(\Delta C_{t-i}, 0), \quad \text{loss}_t = \frac{1}{14} \sum_{i=0}^{13} \max(-\Delta C_{t-i}, 0) \quad (2)$$

where

$$\Delta C_{t-i} = C_{t-i} - C_{t-i-1} \quad (2a)$$

is the one-period price change. Positive values contribute to the gain, while negative values (losses) contribute to the loss. Building on these smoothed components, RSI is then calculated as:

$$\text{RSI}_t = 100 - \frac{100}{1 + \frac{\text{gain}_t}{\text{loss}_t}}. \quad (3)$$

By construction, RSI ranges from 0 to 100; conventional thresholds at 70 (overbought) and 30 (oversold) help flag potential momentum exhaustion. In practice, these levels serve the agent as interpretable signals for tactical mean reversion or trend continuation, depending on the surrounding context.

Trend structure is further summarized using the Simple Moving Average, which compresses recent price information into a low-pass, noise-reduced estimate of the underlying level. The k -day SMA is defined by:

$$\text{SMA}_k(t) = \frac{1}{k} \sum_{i=0}^{k-1} C_{t-i}, \quad k \in \{5, 10\} \quad (4)$$

In this study, two SMAs with lengths of 5 and 10 are employed to detect short-horizon trend cross-overs (SMA crossovers). The relative ordering and intersection of these SMAs provide the agent with compact regime cues—e.g. short-term trend acceleration or deceleration—without introducing excessive lag.

To quantify time-varying dispersion and potential excursion from equilibrium, we adopt Bollinger Bands with a 20-day lookahead and standard deviation 2σ . First, we compute the 20-day moving average:

$$\mu_t = \text{SMA}_{20}(t), \quad (5)$$

followed by the rolling standard deviation of prices over the same 20-day window:

$$\text{std}_t = \sqrt{\frac{1}{20} \sum_{i=0}^{19} (C_{t-i} - \mu_t)^2} \quad (6)$$

The upper and lower bands of the Bollinger construct are then given by:

$$\text{Upper}_t = \mu_t + 2\text{std}_t, \quad \text{Lower}_t = \mu_t - 2\text{std}_t. \quad (7)$$

Intuitively, when price touches the upper or lower Bollinger band, it signals a statistically large deviation from the recent mean. Such excursions often precede partial normalization—useful for mean-reversion tactics—whereas persistent ‘band riding’ can corroborate trend-following dynamics. Taken together, returns, momentum, trend, and volatility envelopes offer a complementary feature set that helps a reinforcement learning (RL) agent infer latent market states and adapt its policy.

In [Figure 2](#), each technical indicator is represented by a distinct color, allowing for intuitive interpretation. The yellow line corresponds to the 5-day SMA, while the orange-red line represents the 10-day SMA; their crossovers reveal short-term momentum shifts. The purple lines form the Bollinger Bands, with the upper, middle, and lower bounds capturing volatility around the 20-day moving average, the

green line denotes the 14-period VWAP, integrating both price and trading volume to indicate the average transaction level.

To illustrate how the previously defined indicators can be applied in practice, consider Figure 2, which depicts a daily candlestick chart of BTC/USDT augmented with three central tools: the Simple Moving Averages (SMA 5 and SMA 10), the Bollinger Bands (20, 2), and the Volume-Weighted Average Price (VWAP, 14). Together, these indicators offer an integrated view of short-term trends, volatility dynamics, and trading pressure, thereby providing a realistic example of how market signals can be distilled into structured features for a reinforcement learning agent.

First, the short-term SMAs serve as a proxy for momentum and trend direction. The SMA(5) responds more quickly to price changes, while the SMA(10) incorporates slightly longer memory. When the faster SMA(5) crosses above the slower SMA(10), traders typically interpret this as a bullish signal, anticipating upward momentum. Conversely, when the SMA(5) falls below the SMA(10), the market often enters a corrective or bearish phase. In the chart under consideration, the SMA(5) has recently slipped below the SMA(10), indicating that short-term corrective forces may be developing despite previous upward momentum.

Second, the Bollinger Bands (20, 2) provide a volatility envelope around the 20-day moving average. The upper and lower bands expand and contract in response to changes in market variability. Prices approaching the upper band often suggest overbought conditions, while proximity to the lower band signals oversold conditions. In this example, Bitcoin's price currently hovers near the central band with a slight tilt toward the lower boundary. This positioning conveys moderate bearish undertones but without evidence of extreme volatility, implying that while corrective forces exist, they remain contained within normal market fluctuations.

Third, the VWAP(14) integrates both price and traded volume to highlight the average level at which the asset has been exchanged over the specified window. Sustained trading above VWAP generally reflects strong buying pressure and bullish sentiment, whereas trading below VWAP is indicative of bearish dominance. At present, BTC lingers beneath the VWAP line, reinforcing the bearish cues suggested by the SMA crossover and the Bollinger configuration. This confluence of signals suggests short-term vulnerability, with the likelihood of consolidation before the market initiates a new directional phase.

Taken collectively, this example illustrates that while each indicator conveys valuable information in isolation, their integration yields a more comprehensive representation of market dynamics. Specifically, the SMA captures directional changes, the Bollinger Bands quantify deviations from equilibrium, and the



Figure 2. BTC/USDT daily candlestick chart with SMA, Bollinger Bands and VWAP (Source: binance.com).

VWAP encapsulates trading consensus weighted by volume. When embedded within the feature space of a reinforcement learning framework, these heterogeneous signals are transformed into structured components of a state vector that compactly characterizes the market environment at each time step. Such a representation enables the agent not merely to detect whether the market is trending, reverting, or consolidating, but also to adapt its decision-making policy in a systematic and data-driven manner.

Accordingly, the constructed feature set possesses an inherently spatial structure: at every temporal juncture, the prevailing market state is encoded within a seven-dimensional vector. This vector constitutes the primary input to the learning model and, once subjected to normalization and/or dimensionality reduction techniques such as Principal Component Analysis, facilitates the agent's ability to discern latent market patterns and interpret trading signals derived from empirical data with enhanced precision and stability.

$$\mathbf{x}_t = [r_t, \text{RSI}_t, \text{SMA}_5(t), \text{SMA}_{10}(t), \text{Upper}_t, \text{Lower}_t, \text{Volume}_t]^\top \in \mathbb{R}^7. \quad (8)$$

3.1.2. Normalization and PCA

Once the technical indicators have been calculated, the next step is to ensure that the resulting features are expressed on a comparable scale. Raw financial variables often differ greatly in magnitude—for example, trading volume can be several orders of magnitude larger than returns or relative indices—leading to instability in training and potential bias in the learning process. To address this issue, all features are normalized so that they lie within the same numerical range, thereby improving model convergence and ensuring that no single variable disproportionately influences the outcome. The Min–Max normalization procedure is adopted for this purpose, and its formula is given as follows:

$$\tilde{x}_{tt} = \frac{\mathbf{x}_t - \mathbf{x}^{\min}}{\mathbf{x}^{\max} - \mathbf{x}^{\min}} \in [0, 1]^7 \quad (9)$$

Where $\mathbf{x}_t$ is the initial feature vector at time t , $\mathbf{x}^{\min}, \mathbf{x}^{\max}$ is the minimum and maximum value vectors in each dimension in the entire **training set**, and $\tilde{\mathbf{x}}_t$ is the feature vector after being normalized, with all elements lying in the interval $[0, 1]$. This normalization **only uses information from the training set**, and then applies the same min/max parameters to the entire testing set.

After normalization, the dataset undergoes dimensionality reduction through the Principal Component Analysis (PCA) method (Greenacre et al., 2022). The objective of this step is to project the normalized seven-dimensional feature vector into a lower, five-dimensional space while still preserving most of the variance—and hence, most of the information—contained in the original data. By doing so, PCA effectively filters out noise, reduces redundancy, and eliminates the high correlations often present among raw features. Mathematically, the transformation is expressed as:

$$\mathbf{z}_t = W^\top \tilde{\mathbf{x}}_t \in \mathbb{R}^5 \quad (10)$$

where $\tilde{\mathbf{x}}_t \in \mathbb{R}^7$ is feature vector after normalization, $W \in \mathbb{R}^{7 \times 5}$ is the projection matrix, containing 5 principal components corresponding to the 5 eigenvectors with the largest variance in the training data and $\mathbf{z}_t \in \mathbb{R}^5$ is the final feature vector (reduced in dimension) fed into the reinforcement learning model. This procedure not only enhances computational efficiency during training but also mitigates the risk of overfitting, ensuring that the model captures the essential structure of the market data without being distracted by irrelevant or redundant fluctuations.

Having transformed and compressed the technical indicators into a reduced feature space, the next step is to embed these features into a reinforcement learning environment where the trading agent can interact with the market. At this stage, the emphasis shifts from preprocessing and representation toward the design of the state, action, and reward components that define the learning problem. This environment forms the backbone of the entire framework, as it governs how the agent perceives information, evaluates outcomes, and adapts its strategy over time.

3.2. Feature selection reinforcement learning environment

The environment serves as the interface between market data and the decision-making process of the agent. Central to this design is the construction of the state, which encodes the most relevant information from recent market history. Specifically, the state at time t is generated by concatenating the reduced feature vectors obtained through PCA over a fixed sliding window. This approach allows the agent to incorporate temporal dependencies and short-term market context into its policy learning, thereby enabling it to recognize patterns such as emerging trends, volatility swings, or potential breakouts.

State: The state at time t is constructed by concatenating the reduced dimensionality feature vectors through PCA over the last $w = 20$ days:

$$s_t = [\mathbf{z}_{t-w+1}^\top, \mathbf{z}_{t-w+2}^\top, \dots, \mathbf{z}_t^\top]^\top \in \mathbb{R}^{5w} = \mathbb{R}^{100}. \quad (11)$$

where $\mathbf{z}_t \in \mathbb{R}^5$ is the feature vector at time t after going through the Min-Max scaling and PCA steps from the original feature. $w = 20$ is the sliding window length, meaning the agent will look at information from the last 20 days before making a decision. Since each day is represented by a 5-dimensional vector, the entire 20-day sequence will form a state vector of size $5 \times 20 = 100$ dimensions. This is the input to the agent in a reinforcement learning environment: a state vector that reflects the short-term market context, allowing the agent to learn about trends, swings, or breakouts in the near-term time frame.

After specifying how the state vector encodes recent market conditions, the next step is to define the **action space**, which governs the set of trading decisions available to the reinforcement learning agent. Unlike a minimal formulation where the agent only chooses between *Buy*, *Sell*, or *Hold*, this study designs a richer action space by embedding predefined technical strategies. In this way, the agent does not merely react at a price level, but instead selects among alternative rule-based strategies grounded in widely used financial indicators. This approach increases interpretability and aligns the decision-making process with established trading heuristics.

3.2.1. Action space

At each time step, the agent can select one out of five possible actions, each corresponding to a predefined technical trading strategy:

$$\mathcal{A} = \{0, 1, 2, 3, 4\}$$

Meaning of each action as follow:

Action 0: RSI-based strategy—generates buy or sell signals when the market enters oversold or overbought conditions, respectively, based on the Relative Strength Index.

Action 1: SMA crossover strategy—compares SMA(5) and SMA(10); a crossover from below signals a bullish regime, whereas a downward crossover suggests a bearish regime.

Action 2: Bollinger Band strategy—interprets touches at the upper or lower band as potential reversal points, indicating mean-reversion opportunities.

Action 3: Momentum-20d strategy—evaluates price momentum by comparing cumulative increases over the last 20 days, thereby capturing medium-term directional strength.

Action 4: VWAP reversion strategy—measures price deviation from the 14-day Volume-Weighted Average Price, assuming that large deviations tend to revert toward the mean.

When the agent chooses an action $a_t \in \mathcal{A}$, the corresponding strategy is applied to calculate the position signal (buy, sell or hold) at that moment.

At each time step t , the reinforcement learning agent selects an action a_t , each corresponding to a specific predefined technical trading strategy. These strategies analyze current market conditions and produce a trading signal σ_t , where $\sigma_t = +1$ indicates opening a long position, $\sigma_t = -1$ indicates opening a short position, and $\sigma_t = 0$ represents holding the current position without change.

To formalize the action space, each trading decision made by the RL agent is explicitly linked to one of these technical strategies, which in turn generate **buy**, **sell**, or **hold** signals based on well-established indicator-based rules. This mapping ensures that the agent's behavior remains interpretable, consistent

with traditional market heuristics, and grounded in systematic decision-making rather than purely data-driven exploration.

Specifically, five technical trading strategies are defined, each corresponding to an action $a_t \in \{0, 1, 2, 3, 4\}$. For each strategy, precise quantitative conditions determine whether a **long**, **short**, or **neutral** position is initiated. The detailed mathematical rules governing the generation of the trading signal σ_t at each time step t are presented below:

$$\begin{aligned}
 0) \text{ RSI Rule: } \sigma_t &= \begin{cases} +1, & \text{if } RSI_t < 30, \\ -1, & \text{if } RSI_t > 70, \\ 0, & \text{otherwise.} \end{cases} \\
 1) \text{ SMA Crossover Rule: } \sigma_t &= \begin{cases} +1, & \text{if } SMA_{5,t} > SMA_{10,t}, \\ -1, & \text{if } SMA_{5,t} < SMA_{10,t}, \\ 0, & \text{if } SMA_{5,t} = SMA_{10,t}. \end{cases} \\
 2) \text{ Bollinger Bands Rule: } \sigma_t &= \begin{cases} +1, & \text{if } C_t < Lower_t, \\ -1, & \text{if } C_t > Upper_t, \\ 0, & \text{if } Lower_t \leq C_t \leq Upper_t. \end{cases} \\
 3) \text{ 20-day Momentum Rule: } \sigma_t &= \begin{cases} +1, & \text{if } C_t > C_{t-20}, \\ -1, & \text{if } C_t < C_{t-20}, \\ 0, & \text{if } C_t = C_{t-20}. \end{cases} \\
 4) \text{ 20-day VWAP Rule: } \sigma_t &= \begin{cases} +1, & \text{if } C_t > VWAP_{20,t}, \\ -1, & \text{if } C_t < VWAP_{20,t}, \\ 0, & \text{if } C_t = VWAP_{20,t}. \end{cases}
 \end{aligned}$$

Each action a_t is not directly a buy or sell instruction but rather a strategic choice for assessing the market through a predefined technical rule. Once the strategy is applied, it produces a discrete trading signal σ_t , which is then used to evaluate performance through the log-return. In this way, the reinforcement learning agent does not attempt to predict exact price levels; instead, it learns to associate particular strategies with favorable or unfavorable outcomes under varying market conditions.

The integration of multiple technical strategies—ranging from trend-following to mean-reversion—within this framework allows the agent to adapt dynamically to different regimes. By repeatedly observing the consequences of its chosen actions, the agent develops a mapping between market context and strategy effectiveness, ultimately enhancing its ability to select the most appropriate trading rule at each point in time.

3.2.2. Reward

The reward function provides the key feedback for the trading agent. In this framework, reward is defined as position-adjusted log-return, ensuring that profits are positive when the agent predicts the correct direction and negative otherwise. By linking rewards directly to trading outcomes, the learning objective naturally aligns with maximizing portfolio growth over time.

The reward R is defined below:

$$R_t = \sigma_t r_t = \sigma_t (\log C_t - \log C_{t-1}). \quad (12)$$

reward R_t is the profit earned at time t , calculated as position-adjusted log-return. If the agent predicts the price direction correctly (long when the price increases, short when the price decreases), R_t it will be positive.

Update account value (NAV, start $V_0 = C_{\text{start}} = 10^6$):

$$V_{t+1} = V_t e^{R_t} = V_0 \exp\left(\sum_{k=0}^t R_k\right). \quad (13)$$

The agent does not learn directly from the price but from the reward, which is a log-return adjusted by the trading direction. The NAV is updated exponentially based on the cumulative total reward. The agent's goal in reinforcement learning is to maximize the total reward, which is equivalent to maximizing the growth of the NAV account over time.

3.3. Reinforcement learning problem

With the state representation, action space, and reward function clearly defined, the trading task can now be formalized as a reinforcement learning problem. In this framework, the market environment is modeled as a Markov Decision Process (MDP), where the agent repeatedly observes the current state, selects an action according to its policy, and receives a reward that reflects trading performance. Over time, the agent aims to learn an optimal policy that maximizes the cumulative expected reward, which in practice corresponds to maximizing portfolio growth while adapting to evolving market dynamics.

In the automated trading model using reinforcement learning, the environment is modeled as a Markov Decision Process, represented as a set of 4 elements:

$$\text{MDP}=(\mathcal{S}, \mathcal{A}, P, R, \gamma) \quad (14)$$

State space $\mathcal{S}$

In this problem, each state $S_t \in \mathbb{R}^{100}$ is a vector of size 100, constructed from the last 20 days of feature data each day is a 5-dimensional vector after PCA. This vector synthesizes short-term technical information about the market to help the agent make decisions.

Action space $\mathcal{A}$

Is a discrete set of 5 actions corresponding to 5 predefined technical strategies: $\mathcal{A} = \{0, 1, 2, 3, 4\}$ The Agent will select one action $a_t \in \mathcal{A}$ at each t moment, corresponding to the application of a specific trading strategy (eg: RSI, SMA, Bollinger ...).

State transition probability function $\mathcal{P}$

Probability representation $P(S_{t+1}|S_t, a_t)$: the probability of transitioning from the current state S_t to the next state S_{t+1} , after performing action a_t .

In financial environments, markets are uncertain and fluctuate over time series so $\mathcal{P}$ they cannot be written explicitly—the agent learns from observed data.

Reward $\mathcal{R}$

$R_t = \sigma_t \cdot (\log C_t - \log C_{t-1})$ as stated in the previous section: is the position-adjusted logarithmic return. This is a signal to assess the quality of the chosen action.

Discount factor γ

In this model, the discount factor is set to $\gamma=0.1$, a deliberately low value chosen to reflect the short-term nature of daily Bitcoin trading. Since each time step corresponds to a single trading day, emphasizing immediate rather than distant rewards is crucial in such a highly volatile and fast-moving market. A low discount factor ensures that the agent prioritizes present gains, capturing short-lived opportunities signaled by technical indicators, while reducing the influence of uncertain long-term forecasts that are often unreliable in cryptocurrency markets. This design not only aligns the agent's objective with short-term profit maximization but also enhances stability by limiting the accumulation of errors in long-horizon return estimates.

Optimal goal: The agent seeks the optimal policy $\pi(a|s)$, i.e. the mapping function from state to action probability, such that the cumulative expected value function of the total reward is maximized:

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right]. \quad (15)$$

Where $J(\pi)$ is the total expected reward under policy π , T is the total number of time steps in an episode, $\gamma^t R_t$ the reward at step t , discounted over time. The $\mathbb{E}_{\pi}$ expectation notation is taken from the action distribution generated by policy π .

3.4. DQN algorithm

The following section details the implementation of a Deep Q-Learning algorithm specifically designed for Bitcoin trading tasks. This approach utilizes a multilayer perceptron (MLP) to approximate the value-action function and is trained via temporal difference (TD) learning. Through the use of a replay buffer and epsilon-greedy exploration strategy, the model aims to learn an optimal trading policy by interacting with a simulated environment under a set of predefined technical strategies.

Approximate value action function:

$$Q_\theta : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}, \quad Q_\theta(s, a) \approx \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k} | s_t = s, a_t = a \right], \quad (16)$$

modeled by MLP network.

Regression target (TD standard):

$$y_t = R_t + \gamma \max_{a'} Q_{\theta^-}(s_{t+1}, a'), \quad (17)$$

and loss

$$L(\theta) = (y_t - Q_\theta(s_t, a_t))^2 \quad (18)$$

with θ^- is the ‘target network’ parameter (slow update).

The algorithm performs gradient descent above $L(\theta)$ with random batch taken from replay buffer $\mathcal{D}$ (size 50,000).

Exploration strategy ε – greedy:

$$\varepsilon(t) = \begin{cases} \varepsilon_{\text{init}} - (\varepsilon_{\text{init}} - \varepsilon_{\text{final}}) \cdot \frac{t}{N \cdot f} & \text{if } t \leq N \cdot f, \\ \varepsilon_{\text{final}} & \text{if } t > N \cdot f \end{cases}. \quad (19)$$

where $\varepsilon(t)$ is the epsilon value at time step t , which controls the exploration rate; $\varepsilon_{\text{init}} = 1$ is the initial epsilon value, meaning the agent explores almost randomly at the beginning; $\varepsilon_{\text{final}} = 0.06$ is the final epsilon value, meaning the agent exploits learned policies more often later; N is the total number of episodes; f is a scaling factor or update frequency, t is the current training step or time index.

3.4.1. Key hyperparameters

The training process relies on carefully chosen hyperparameters to balance stability and learning efficiency. In this model, the learning rate is set to $\eta = 5 \times 10^{-5}$, ensuring gradual updates to the neural network weights and reducing the risk of divergence in a volatile trading environment. The total number of training steps is fixed at $N_{\text{steps}} = 200\,000$, providing sufficient iterations for the agent to explore the state–action space and converge toward an effective trading policy.

Algorithm 1 is designed to optimize the trading policy through iterative interaction with the market environment, learning from historical experiences, and balancing the trade-off between exploration and exploitation using an epsilon-greedy strategy.

The complete procedural steps, including initialization, experience replay, temporal difference target computation, and periodic updates of both the main and target networks, are summarized in **Algorithm 1: Deep Q-Learning for Bitcoin Transactions**. The detailed pseudo-code and implementation outline are provided in **Appendix** for reference and reproducibility.

Algorithm 1. Deep Q-Learning for Bitcoin Transactions

Input:

Number of training steps T
 Replay buffer $\mathcal{D}$ (maximum size $N = 50\,000$)
 Trading environment $\mathcal{E}$
 Discount factor $\gamma = 0.1$
 Batch size: B
 Main Q network Q_θ , and **target network** $Q_{\theta^-} \leftarrow Q_\theta$

Initialization:

Initialize random network parameters θ
 Replay buffer $\mathcal{D} \leftarrow \emptyset$
 Set initial status $s_0 \in \mathbb{R}^{100}$ from last 20 days

For $t = 0$ to $T - 1$ **do:**

1. **Select action a** according to the epsilon-greedy policy:

With probability ε : choose a random action $a_t \in \{0, 1, 2, 3, 4\}$
 opposite: $a_t = \arg\max_a Q_\theta(s_t, a)$

2. Apply the corresponding strategy $a_t \rightarrow$ to generate position signals $\sigma_t \in \{-1, 0, +1\}$

3. Get Rewards:

$$R_t = \sigma_t \cdot \log \left(\frac{C_t}{C_{t-1}} \right)$$

4. Move to next state s_{t+1} (sliding window)

5. Save experience to buffer:

$$\mathcal{D} \leftarrow \mathcal{D} \cup \{(s_t, a_t, R_t, s_{t+1})\}$$

6. Take a random minibatch of BBB elements from $\mathcal{D}$

7. Calculate TD target for each sample in the batch:

$$y_i = R_i + \gamma \cdot \max_{a'} Q_{\theta^-}(s_{i+1}, a')$$

8. Calculate the loss function:

$$L(\theta) = \frac{1}{B} \sum_{i=1}^B (y_i - Q_{\theta}(s_i, a_i))^2$$

9. Update the main network parameter θ using gradient descent:

$$\theta \leftarrow \theta - \eta \cdot \nabla_{\theta} L(\theta)$$

10. Periodically update target network:

o If $t \bmod \tau = 0$ (e.g. every 1000 steps), then:

$$\theta^- \leftarrow \theta$$

End For

Output:

Optimal policy $\pi(a|s) = \operatorname{argmax}_a Q_{\theta}(s, a)$

The trained model can be applied to predict and trade on new data.

Test phase

Once training is complete, **the policy determines**

$$\pi^*(s) = \operatorname{argmax}_a Q_{\theta^*}(s, a) \quad (20)$$

is applied sequentially on the test dataset to generate pairs (t, σ_t, V_t) . Upon completion of the training process, the learned policy is evaluated on an out-of-sample test dataset to examine its generalization capability. The trained agent sequentially applies the optimized policy function (20) across the test period, thereby generating a series of trading signals and updating the Net Asset Value trajectory. This evaluation process produces both quantitative and visual performance outputs: a structured record of signals, NAV, and corresponding strategies stored in an excel file (time, signal, NAV, strategy), as well as a graphical representation of NAV progression accompanied by annotated buy and sell points.

This phase serves as a critical validation step, ensuring that the agent's decision-making extends beyond the training environment and remains effective under unseen market conditions. By systematically monitoring trading outcomes and capital growth, the test phase provides empirical evidence for the robustness and practical viability of deep reinforcement learning in algorithmic Bitcoin trading.

Following the design of the reinforcement learning environment and the training of the agent, the next stage is to evaluate the empirical performance of the proposed framework. This evaluation not only verifies the effectiveness of the preprocessing and modeling steps but also provides evidence of the agent's capacity to generalize beyond the training period. Accordingly, the following section presents the experimental results, beginning with a description of the dataset used and the train-test split strategy, followed by an analysis of the agent's performance in the test phase.

4. Results

The results presented in this section provide the empirical foundation for assessing the practical viability of reinforcement learning in Bitcoin trading. By grounding the evaluation in long-term historical data and testing on recent market conditions, the analysis ensures that performance is measured under realistic scenarios rather than overfitting to past patterns. The next section builds upon these findings by discussing the model's behavior under different market regimes, its robustness to volatility, and its implications for real-world algorithmic trading applications.

4.1. Usage data

The data used in the study is the Bitcoin price time series with the trading code 'BTC-USD', collected from Yahoo Finance via the `yfinance` library. The collection period spans from January 1, 2010 to May 21, 2025, including both the training and testing phases of the model. The original data includes the basic fields in the OHLCV format: Open price, High price, Low price, Close price, and Volume. Based on that, the technical features are calculated to serve as inputs for the reinforcement learning model, including: log-return, RSI with a period of 14 days, simple moving averages SMA 5 and SMA 10, Bollinger Bands with a period of 20 days and standard deviation 2σ , along with VWAP over the last 20 days. These features help the model capture the momentum, trend, volatility and overbought/oversold levels of the market at each point in time.

4.1.1. Split train/test set

The dataset is split into two parts (Table 1): a training set that includes all data from January 1, 2012 to December 31, 2021, and a test set that includes data from January 1, 2022 to May 21, 2025. This split is done intentionally to ensure that the model can learn long-term time series features during the training phase, thereby capturing market behavior over multiple cycles. At the same time, the test set is kept intact for the most recent period to test the model's generalization ability and evaluate its true performance when applied to data that has never been seen during training.

4.2. Training results

Figure 3 shows the training process of the Deep Q-Network reinforcement learning model, through technical indicators at two different points in the learning process. It can be seen that the average reward per episode (`ep_reward_mean`) has increased from 29.5 to 30.6, while the loss function has decreased significantly from 0.000646 to 0.000291. This shows that the model is gradually converging and improving its trading policy over time. Parameters such as the number of weight updates (`n_updates`), total time steps (`total_timesteps`), and number of episodes all increase steadily, reflecting that the training process is stable. At the same time, the exploration rate has decreased to the minimum (0.06), indicating that the agent has almost stopped exploring and is focusing on exploiting the learned policy.

4.3. Investment order performance on in-sample and out-of-sample test sets

Over the test window **2022-01-01** → **2025-06-13** (with **2025-05-22** → **2025-06-13** held strictly out-of-sample), the agent executed **1,240 trades** starting from $V_0 = \$1,000,000$. As shown in Figure 4, the final NAV reached **\$125,903,751.6**, which is **125.90×** the initial capital (a cumulative return of **+12,490.38%**). This is greater than $120\times$ expansion evidences strong generalization: the policy adapts across regimes, combining trend-following and mean-reversion rules to compound gains under highly volatile Bitcoin conditions.

Table 2 reports the sequence of trading actions, signals, and strategies applied by the reinforcement learning agent during the test phase. Each row corresponds to a trading day, showing the executed action, the resulting buy/sell signal, the selected technical strategy, and the updated Net Asset Value.

The early period of testing (January–February 2022) highlights the agent's ability to switch dynamically between strategies such as SMA-Crossover, Bollinger Reversion, RSI, and Momentum-20d, depending

Table 1. Data used for training and testing.

TRAIN DATA						TEST DATA					
Time	Open	High	Low	Close	Volume	Time	Open	High	Low	Close	Volume
2012-01-01 00:00:00	4.722	5.5	4.615	5.268	108509.2	2022-01-01 00:00:00	46197.31	47950.29	46196.03	47737.35	16697.92
2012-01-02 00:00:00	5.268	5.47	4.8	5.217	69150.93	2022-01-02 00:00:00	47737.35	47979.26	46666.18	47306.41	13025.66
2012-01-03 00:00:00	5.217	5.29	4.65	4.881	125170.3	2022-01-03 00:00:00	47306.41	47572.23	45707.55	46450.79	21681.38
2012-01-04 00:00:00	4.881	5.7	4.751	5.574	131170.9	2022-01-04 00:00:00	46450.79	47529.48	45587.91	45824.54	29035.93
2012-01-05 00:00:00	5.574	7.22	5.574	6.948	182328.2	2022-01-05 00:00:00	45824.54	47067.18	42510.86	43432.51	47769.2
2012-01-06 00:00:00	6.948	7.21	6.126	6.697	218077.5	2022-01-06 00:00:00	43432.51	43779.23	42468.76	43094.54	36804.78
2012-01-07 00:00:00	6.697	7.016	6.38	6.81	48387.58	2022-01-07 00:00:00	43094.54	43138.79	40750.42	41543.79	51310.46
2012-01-08 00:00:00	6.81	7.2	6.734	7.114	57193.53	2022-01-08 00:00:00	41543.79	42306.77	40534.37	41686.67	27559.19
2012-01-09 00:00:00	7.114	7.191	5.81	6.326	148406.9	2022-01-09 00:00:00	41686.67	42800.66	41237.13	41869.6	21486.68
2012-01-10 00:00:00	6.326	6.89	6.001	6.362	110210.3	2022-01-10 00:00:00	41869.6	42251.07	39667.6	41832.59	51976.97
2012-01-11 00:00:00	6.362	7.138	6.362	6.9	99442.07	2022-01-11 00:00:00	41832.59	43122.32	41279.45	42743.03	36554.37
2012-01-12 00:00:00	6.9	6.997	6.337	6.8	140116.5	2022-01-12 00:00:00	42743.03	44293.83	42471.08	43919.67	31859.05
2012-01-13 00:00:00	6.8	6.804	6.4	6.41	60228.77	2022-01-13 00:00:00	43919.67	44442.67	42334.75	42575.61	29159.4
2012-01-14 00:00:00	6.41	6.95	6.22	6.75	141896.3	2022-01-14 00:00:00	42575.61	43467.05	41785.44	43089.83	27290.3
2012-01-15 00:00:00	6.75	7.08	6.741	7.002	62233.33	2022-01-15 00:00:00	43089.83	43811.94	42589.82	43088.74	14075.12
2012-01-16 00:00:00	7.002	7.189	6.511	6.683	122666.6	2022-01-16 00:00:00	43088.74	43484.21	42617.07	43100.36	12615.04
2012-01-17 00:00:00	6.683	6.95	4.64	5.6	213409.8	2022-01-17 00:00:00	43100.36	43198.05	41571.67	42224.38	20310.91
2012-01-18 00:00:00	5.6	6.95	5.1	5.92	226681	2022-01-18 00:00:00	42224.38	42682.77	41301.01	42372.75	26287.26
2012-01-19 00:00:00	5.92	6.361	5.873	6.36	114820.5	2022-01-19 00:00:00	42372.75	42575.28	41144.83	41674.8	32436.62
2012-01-20 00:00:00	6.36	6.58	6.258	6.49	66786.48	2022-01-20 00:00:00	41674.8	43513.14	40564.32	40697.03	35135.79
2012-01-21 00:00:00	6.49	6.548	6.1	6.18	64761.69	2022-01-21 00:00:00	40697.03	41116.66	35494.42	36469.3	94195.63
2012-01-22 00:00:00	6.18	6.455	6.1	6.31	44890.65	2022-01-22 00:00:00	36469.3	36811.86	34130.35	35075.92	68447.95
2012-01-23 00:00:00	6.31	6.389	6.212	6.356	17693.05	2022-01-23 00:00:00	35075.92	36539.19	34648.38	36288.15	39599.36
2012-01-24 00:00:00	6.356	6.394	6.219	6.29	35663.08	2022-01-24 00:00:00	36288.15	37469.19	32990.72	36702.79	94597.7
2012-01-25 00:00:00	6.29	6.322	5.502	5.756	133115.3	2022-01-25 00:00:00	36702.79	37555.35	35732.05	36975.99	44562.17
2012-01-26 00:00:00	5.756	5.92	5.3	5.34	123954.1	2022-01-26 00:00:00	36975.99	38930.36	36275.92	36829.37	57460
2012-01-27 00:00:00	5.34	5.51	5.05	5.292	115567.8	2022-01-27 00:00:00	36829.37	37242.79	35543.94	37189.85	46006.99
2012-01-28 00:00:00	5.292	5.76	5.26	5.627	89708.02	2022-01-28 00:00:00	37189.85	38021.06	36176.78	37744.7	38853.31
2012-01-29 00:00:00	5.627	5.7	5.328	5.382	46430.45	2022-01-29 00:00:00	37744.7	38728.29	37343.45	38184.86	24097.29
2012-01-30 00:00:00	5.382	5.6	5.367	5.49	42303.14	2022-01-30 00:00:00	38184.86	38364.51	37377.53	37908.44	17354.8
2012-01-31 00:00:00	5.49	5.65	5.426	5.484	48296.66	2022-01-31 00:00:00	37908.44	38769.81	36660.7	38494.66	34716.41

We include the out-of-sample data provided in the attached file 'BITCOIN_daily_data_23_09_2025.xlsx'.

rollout/	
ep_len_mean	3.62e+03
ep_rew_mean	30.6
exploration_rate	0.06
time/	
episodes	52
fps	555
time_elapsed	338
total_timesteps	187980
train/	
learning_rate	5e-05
loss	0.000291
n_updates	46969

Figure 3. DQN model training log—reward, loss and training progress over time.

on prevailing market conditions. For example, in late January 2022 the agent alternated between SMA-based signals and Bollinger-based reversion rules, reflecting short-term corrective behavior in the Bitcoin market. During this stage, NAV fluctuated within a relatively narrow range (around \$930,000–\$1,170,000), suggesting an exploratory phase where the policy balanced between different strategies without committing to a dominant trend.

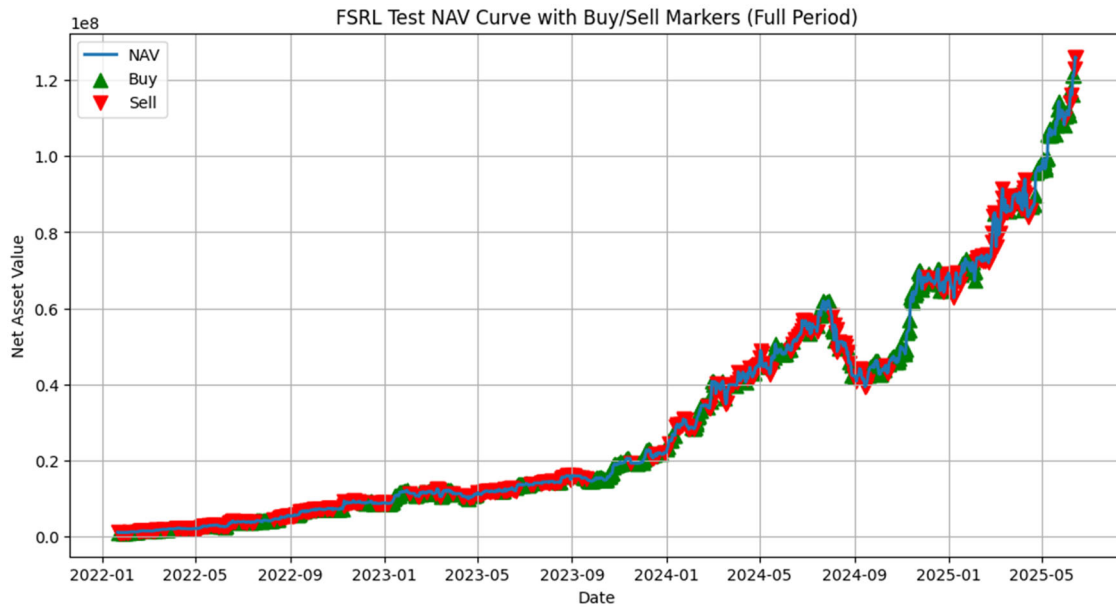


Figure 4. NAV growth chart of reinforcement learning agent on test set (2022–2025) with buy/sell score.

Table 2. Transaction history details of reinforcement learning agent: actions, strategies, and NAV values on test set (the detailed results from January 21, 2022 to June 14, 2025 are provided in the attached file ‘results from 22-1-2022 to 14-06-2025.xlsx’).

Time	Action	Signal	Strategy	Nav	Time	Action	Signal	Strategy	Nav
2022-01-21 00:00:00	1	−1	SMA-Crossover	1025662	2025-05-25 00:00:00	3	1	Momentum-20d	110479578.4
2022-01-22 00:00:00	2	1	Bollinger-Reversion	919112.4	2025-05-26 00:00:00	3	1	Momentum-20d	111731422.8
2022-01-23 00:00:00	1	−1	SMA-Crossover	956462.7	2025-05-27 00:00:00	3	1	Momentum-20d	112158394.8
2022-01-24 00:00:00	2	0	Bollinger-Reversion	956462.7	2025-05-28 00:00:00	3	1	Momentum-20d	111699028.4
2022-01-25 00:00:00	3	−1	Momentum-20d	946571.3	2025-05-29 00:00:00	3	1	Momentum-20d	110472708.8
2022-01-26 00:00:00	3	−1	Momentum-20d	938996	2025-05-30 00:00:00	3	1	Momentum-20d	108269969.4
2022-01-27 00:00:00	4	1	VWAP-Reversion	936235.3	2025-05-31 00:00:00	3	−1	Momentum-20d	109989354.9
2022-01-28 00:00:00	0	1	RSI-Strategy	943517.6	2025-06-01 00:00:00	3	1	Momentum-20d	110668973.6
2022-01-29 00:00:00	1	−1	SMA-Crossover	927228.1	2025-06-02 00:00:00	3	1	Momentum-20d	111739854.3
2022-01-30 00:00:00	4	1	VWAP-Reversion	936338.3	2025-06-03 00:00:00	3	1	Momentum-20d	111992275.8
2022-01-31 00:00:00	0	1	RSI-Strategy	930653.4	2025-06-04 00:00:00	3	1	Momentum-20d	111511989.9
2022-02-01 00:00:00	0	0	RSI-Strategy	930653.4	2025-06-05 00:00:00	3	1	Momentum-20d	110788999.6
2022-02-02 00:00:00	0	0	RSI-Strategy	930653.4	2025-06-06 00:00:00	3	−1	Momentum-20d	114253571.9
2022-02-03 00:00:00	3	−1	Momentum-20d	975968.6	2025-06-07 00:00:00	3	1	Momentum-20d	117421400.5
2022-02-04 00:00:00	1	1	SMA-Crossover	981369.9	2025-06-08 00:00:00	3	−1	Momentum-20d	116057426.4
2022-02-05 00:00:00	1	1	SMA-Crossover	1096342	2025-06-09 00:00:00	3	1	Momentum-20d	116250291.2
2022-02-06 00:00:00	3	−1	Momentum-20d	1097939	2025-06-10 00:00:00	3	1	Momentum-20d	121198027.7
2022-02-07 00:00:00	3	1	Momentum-20d	1123523	2025-06-11 00:00:00	3	1	Momentum-20d	121160787.1
2022-02-08 00:00:00	3	1	Momentum-20d	1161883	2025-06-12 00:00:00	3	−1	Momentum-20d	122917548.4
2022-02-09 00:00:00	3	1	Momentum-20d	1168296	2025-06-13 00:00:00	3	−1	Momentum-20d	126121853.5
2022-02-10 00:00:00	3	1	Momentum-20d	1174948	2025-06-14 00:00:00	3	−1	Momentum-20d	125903751.6

By contrast, the later period (May–June 2025) shows a pronounced reliance on the Momentum-20d strategy, with the agent consistently generating long signals as Bitcoin entered an extended upward trajectory. This concentration of actions coincided with exponential NAV growth: from around \$11 million on May 25, 2025, the account value surged to \$125.9 million by June 14, 2025. This corresponds to a 125.9-fold increase relative to the initial capital of \$1 million, underscoring the agent’s ability to capture sustained bullish momentum in the out-of-sample period.

During the out-of-sample test from July 1, 2025 to September 23, 2025, the trained agent exhibited a strong ability to adapt to new market conditions, as illustrated in Figure 5. The Net Asset Value trajectory reflects the critical influence of position alignment with market trends: when the agent’s strategies were consistent with the prevailing direction (long during upswings, short during downswings), NAV increased steadily and compounded returns over time. Conversely, when positions were taken against the dominant trend, temporary drawdowns occurred, producing noticeable downward movements in the NAV curve. This alternating sequence of gains and corrections highlights the inherent risk–reward

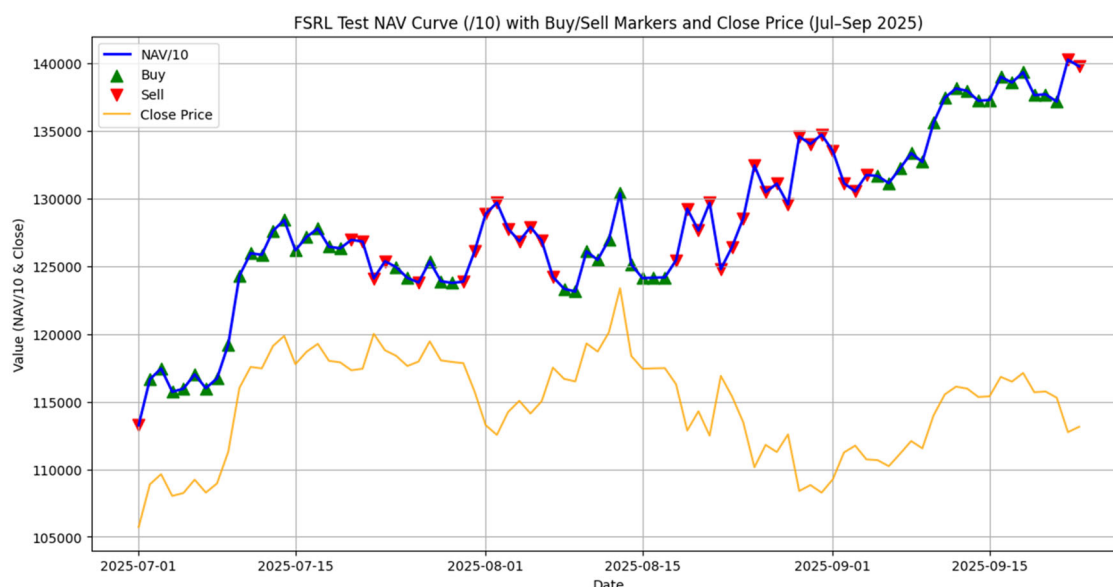


Figure 5. Out-of-sample NAV performance of the trained agent from 2025-07-01 to 2025-09-23, illustrating aligned versus misaligned trading positions.

balance of reinforcement learning-based trading. Importantly, the overall upward trajectory demonstrates that the cumulative gains from aligned trades substantially outweighed losses from misaligned ones, confirming the robustness and profitability of the agent's decision-making even in previously unseen market environments.

Before moving to the conclusion, it is useful to briefly synthesize the insights gained from the methodology and results. The study demonstrates that combining well-established technical indicators with reinforcement learning provides a structured way to interpret market dynamics and generate adaptive trading strategies. The experimental results—both in-sample and out-of-sample—show that the agent can effectively exploit short-term opportunities while also adapting to rapidly changing market conditions. These findings highlight not only the robustness of the approach but also its potential applicability in real-world algorithmic trading, setting the stage for broader implications and future research directions that will be discussed in the conclusion.

5. Conclusion

This study proposed and implemented a Deep Q-Network-based reinforcement learning framework in which the action space is redefined as a set of five predefined technical trading strategies (RSI, SMA Crossover, Bollinger Bands, Momentum-20d, VWAP Reversion). By leveraging both domain-specific expert knowledge and the adaptive capacity of reinforcement learning, the model overcomes the challenges of non-stationarity and high volatility in Bitcoin markets.

The empirical evaluation confirms the effectiveness of this approach. In the main test period (2022–2025), the trained agent achieved exceptional performance, with NAV growing more than 120 times the initial capital, as shown in Figure 4. The results further indicate that the agent adapts flexibly across different market regimes: prioritizing momentum-based strategies during strong trends, and selectively applying mean-reversion or RSI-based strategies when markets enter corrective phases. The out-of-sample test (July–September 2025, Figure 5) reinforces this conclusion, where aligned positions with market direction drove NAV upward, while misaligned positions caused temporary drawdowns. Nevertheless, cumulative gains from correct alignment far outweighed losses, underscoring the robustness and generalization capacity of the learned policy.

From a theoretical perspective, the findings support the Adaptive Markets Hypothesis, showing how AI-driven agents can adjust strategy selection in response to evolving market efficiency. The integration of technical rules within the action space also reflects behavioral finance insights, linking strategy

effectiveness to investor psychology under different conditions. This synergy between reinforcement learning and financial theory enhances both the robustness and interpretability of the model.

The framework offers several practical advantages: increased explainability through rule-based actions, reduced risk of overfitting via dimensionality reduction and feature selection, and strong extensibility toward multi-asset or multi-strategy applications. Future work may expand the framework by incorporating qualitative signals (news, sentiment), adopting multi-objective learning to balance risk and return, or extending the agent to portfolios of digital assets.

Finally, beyond its technical contributions, this research has policy implications for emerging markets such as Vietnam. The rapid adoption of AI in algorithmic trading highlights the urgency of developing appropriate regulatory frameworks to ensure transparency, investor protection, and market integrity. Building safe fintech infrastructure, fostering AI talent in finance, and aligning with international standards on AI governance will be essential to balancing innovation with stability in the evolving digital financial ecosystem.

Author contributions

CRedit: **Nguyen Thi Thu Hoan**: Data curation, Formal analysis, Investigation, Supervision, Writing – review & editing; **Nguyen Ngoc Khang**: Resources, Software, Visualization; **Pham Van Khanh**: Conceptualization, Methodology, Writing – original draft.

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Data availability statement

We hereby confirm that the data supporting the findings of this research paper are derived from public domain resources. We agree to make the data and materials supporting the results and analyses presented in our paper

available upon reasonable request. All source code and datasets used in this study are available at: <https://github.com/khanhvietdm-glitch/BTC-Trading>

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